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CALCULATION AND PLOTTING 3D FIGURES OF VARIABLE X4SU FOR PARAMETER Ssu

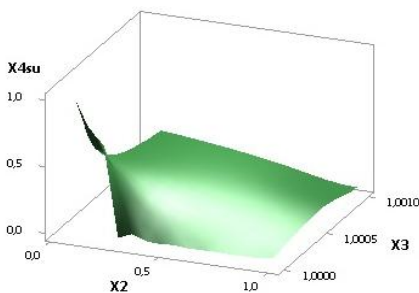
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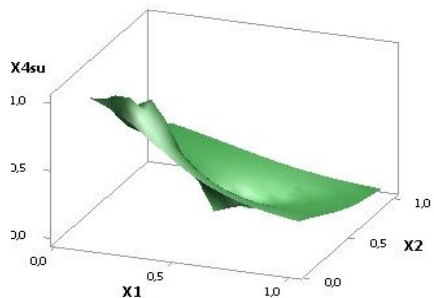
Abstract. Here is reviewed a practical issue of making calculations to obtain a value of the X4su variable, to which end are used X1, X2, X3 variables and the Ssu parameter. The values of the X4su variable obtained allow to identify limits within which they can exist.

Keywords: X4su calculated variable, GDP-defining Ssu parameter, 3D graphs, Excel.

The author had previously carried out calculations of the volume of the economic shell S, described in the older articles. The article below shows how the values of three variables X1, X2 and X3 and of the parameter Ssu (GDP) affect calculations of the variable X4su and plotting of 3D graphs. In this case, the values of the variables may remain constant, increase or decrease by a factor of 10. That is to say the changes of $X4su = f(X1, X2, X3, Ssu)$ are being evaluated.



**Fig. 1. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = X3 = Ssu = 1, X2 = 1...0,1$**



**Fig. 2. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = X2 = 1...0,1, X3 = Ssu = 1$**

Thus, Figure 1 shows a 3D graph of the X4su dependence, with the variables having the following values $X1 = X3 = Ssu = 1, X2 = 1...1$. As seen in this figure, the X4su values increase by a factor of 1.2. The next Figure 2 shows a 3D graph for X4su, which increases by a factor of 2.7 with the following variables $X1 = X2 = 1...0,1, X3 = Ssu = 1$.

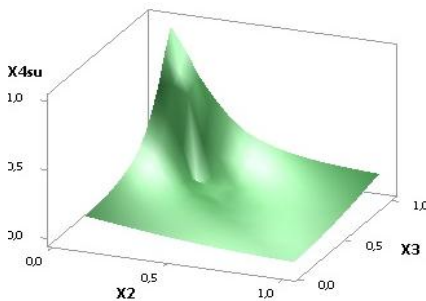


Fig. 3. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = 1, X2 = X3 = Ssu = 0,1...1$

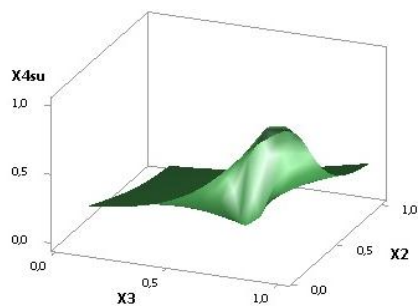


Fig. 4. $X4su = f(X1, X2, X3, Ssu)$
 $X1=X2=1...0,1,X3= Ssu= 0,1...1$

The following Figures 3 and 4 represent two 3D graphs for $X4su$, with the variables being $X1 = 1, X2 = X3 = Ssu = 0,1...1$ and $X1 = X2 = 1...0,1, X3 = Ssu = 0,1...1$ respectively. As seen in Figure 3, the plotted $X4su$ dependence increases by a factor of 1.4. In its turn, 3D graph in Figure 4 increases by a factor of 1.26.

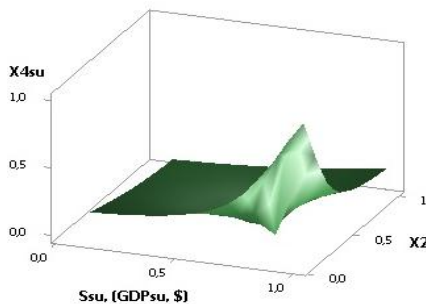


Fig. 5. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = X2 = X3 = 1...0,1, Ssu = 0,1...1$

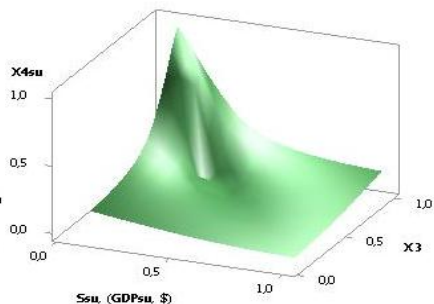
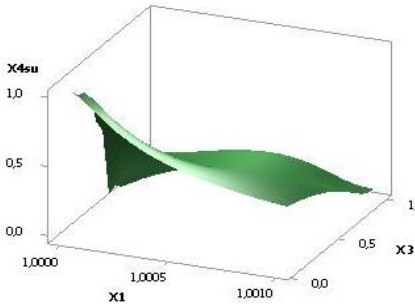
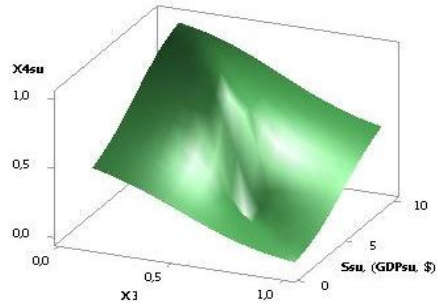


Fig. 6. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = X2 = Ssu = 1...0,1, X3 = 0,1...1$

The values calculated for the $X4su$ dependence in Figure 5 with variables $X1 = X2 = X3 = 1...0,1, Ssu = 0,1...1$ increase by a factor of 1.29. The $X4su$ values for a 3D graph in Figure 6 also increase by a factor of 1.29 with $X1 = X2 = Ssu = 1...0,1, X3 = 0,1...1$.

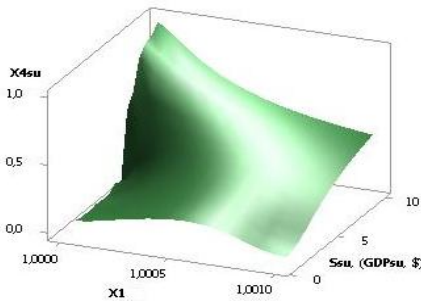


**Fig. 7. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = 1, X2 = X3 = 1...0,1, Ssu = 1...10$**

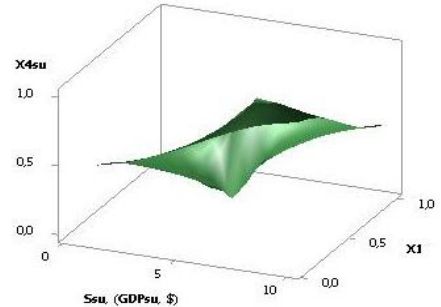


**Fig. 8. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = X2 = X3 = 1...0,1, Ssu = 1...10$**

The 3D graphs in Figures 7 and 8 were plotted with $X1 = 1, X2 = X3 = 1...0,1, Ssu = 1...10$ and $X1 = X2 = X3 = 1...0,1, Ssu = 1...10$ respectively. Here in Figure 7 the values of the $X4su$ variable increase by a factor of 1.77, while in Figure 8 increase by a factor of 1.26.



**Fig. 9. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = 1, X2 = X3 = 1...0,1, Ssu = 1...10$**



**Fig. 10. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = X2 = X3 = 1...0,1, Ssu = 1...10$**

The next 3D graphs in Figures 9 and 10 were plotted with $X1 = 1, X2 = X3 = 1...0,1, Ssu = 1...10$ and $X1 = X2 = X3 = 1...0,1, Ssu = 1...10$ respectively. Here, in Figure 9, $X4su$ increases by a factor of 1.77. The values in Figure 10 increase by a factor of 1.26.

Figure 11 shows that the 3D graph plotted for $X4su$ with $X1 = 1, X2 = 1...0,1, X3 = 0,1...1, Ssu = 1...10$ increases by a factor of 1.14. Figure 12 shows that the 3D graph for $X4su$ with variables $X1 = X2 = 1...0,1, X3 = 0,1...1, Ssu = 1...10$ increases by a factor of 1.16.

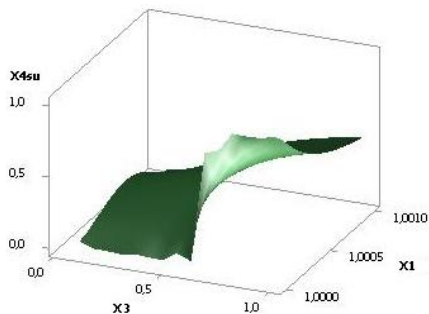


Fig. 11. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = 1, X2 = 1...0,1, X3 = 0,1...1,$
 $Ssu = 1...10$

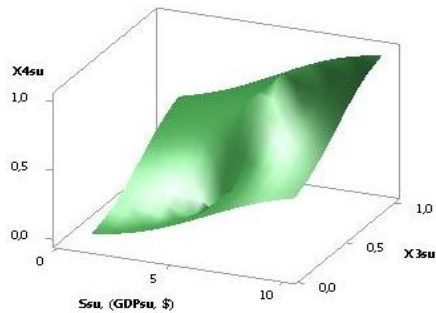


Fig. 12. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = X2 = 1...0,1, X3 = 0,1...1, Ssu = 1...10$

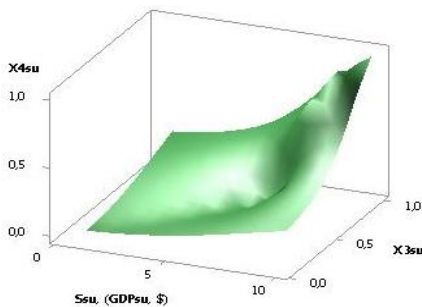


Fig. 13. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = 1...0,1, X2 = 1, X3 = 0,1...1,$
 $Ssu = 1...10$

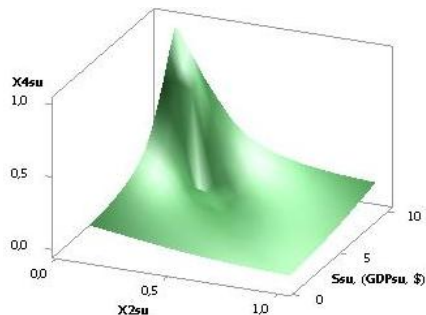


Fig. 14. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = Ssu = 1...10, X2 = 1...0,1,$
 $X3 = 0,1...1$

Figure 13 shows the 3D graph for $X4su$ with $X1 = 1...0,1, X2 = 1, X3 = 0,1...1, Ssu = 1...10$, increasing by a factor of 1.35. 3D graph for $X4su$, depicted in Figure 14, clearly increases by a factor of 1.29. This graph was plotted with the following values of the variables $X1 = Ssu = 1...10, X2 = 1...0,1, X3 = 0,1...1$.

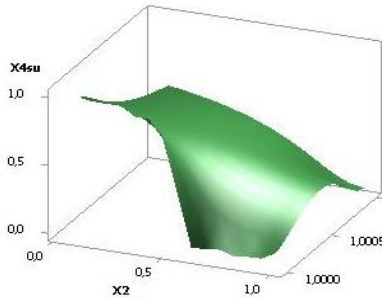


Fig. 15. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = X3 = 1, X2 = 1...0,1, Ssu = 1...10$

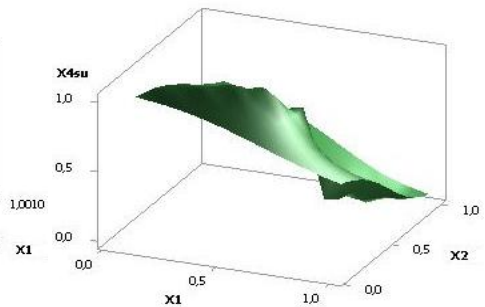


Fig. 16. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = X2 = 1...0,1, X3 = 1, Ssu = 1...10$

The next 3D graph in Figure 15 was plotted with the variables $X1 = X3 = 1, X2 = 1...0,1, Ssu = 1...10$. Here the 3D graph increases by a factor of 1.26. The following variables $X1 = X2 = 1...0,1, X3 = 1, Ssu = 1...10$ were used to plot a 3D graph in Figure 16. The resulting 3D graph for $X4su$ increases by a factor of 1.28.

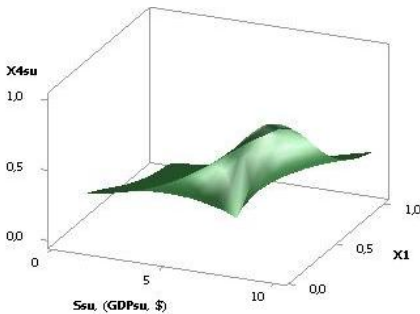


Fig. 17. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = 1...0,1, X2 = X3 = 1, Ssu = 1...10$

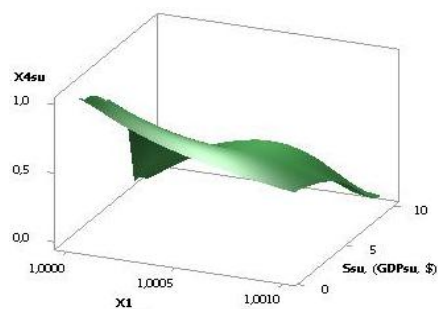


Fig. 18. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = X3 = 1, X2 = 1...0,1, Ssu = 10...1$

Figure 17 shows a 3D graph for $X4su$ with $X1 = 1...0,1, X2 = X3 = 1, Ssu = 1...10$, demonstrating that $X4su$ increases by a factor of 1.77. The 3D graph for $X4su$ in Figure 18, with variables $X1 = X3 = 1, X2 = 1...0,1, Ssu = 10...1$, increases by a factor of 2.01.

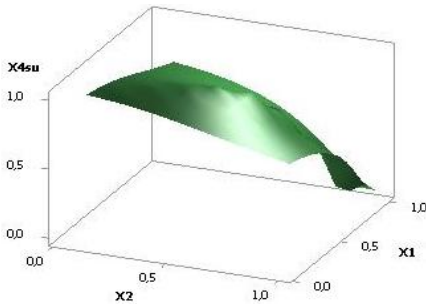


Fig. 19. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = X2 = 1...0,1, X3 = 1...0,1,$
 $Ssu = 10...1$

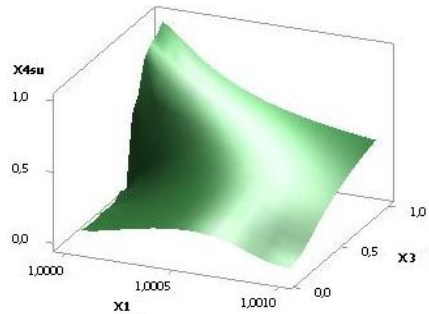


Fig. 20. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = 1, X2 = 1...0,1, X3 = 0,1...1,$
 $Ssu = 10...1$

The 3D graph of $X4su$ in Figure 19 increases by a factor of 2.7 with variables $X1 = X2 = 1...0,1, X3 = 1...0,1, Ssu = 10...1$. The 3D graph presented for $X4su$ in Figure 20 increases by a factor of 1.77 with $X1 = 1, X2 = 1...0,1, X3 = 0,1...1, Ssu = 10...1$.

The next Figure 21 shows the values of $X4su$ increasing by a factor of 1.26 with $X1 = 1, X2 = 1...0,1, X3 = 0,1...1, Ssu = 10...1$. As seen in Figure 22, the 3D graph increases from 0.49 to 0.989, i.e. by a factor of 2.01 with the variables $X1 = X2 = X3 = 1...0,1, Ssu = 10...1$.

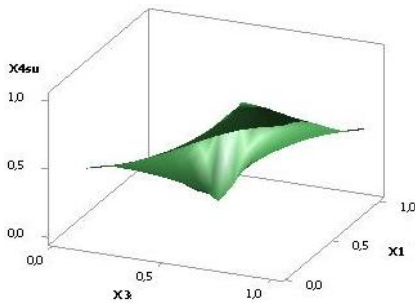


Fig. 21. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = X2 = 1...0,1, X3 = 0,1...1,$
 $Ssu = 10...1$

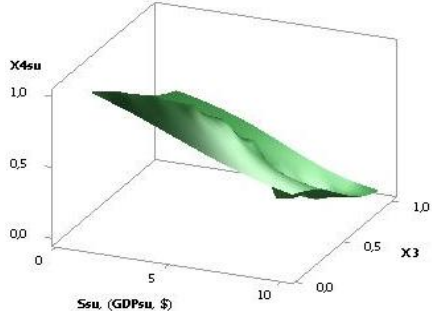


Fig. 22. $X4su = f(X1, X2, X3, Ssu)$
 $X1 = X2 = X3 = 1...0,1,$
 $Ssu = 10...1$

The calculations made form a basis for the Combined Table 1, which includes both the calculated values of the $X4su$ variable and the other values used in calculations. Here the variables $X4suf$ and $X4suf$ signify the starting and final values of the $X4su$ variable and indicate by which factor the $X4su$ value did increase or decrease. This table includes calculations with $X4suf / X1sub \geq 1$.

Table 1 – Arranging ratios of the $X4_{\text{suf}} / X4_{\text{sub}}$ parameters in descending order

№ п/п	X1 ед.	X2 ед.	X3 ед.	$X4_{\text{suf}}$ ед.	$S_{\text{suf}} \text{ ед.}^2$ (GDP_{suf} , \$)	$X4_{\text{suf}} / X4_{\text{sub}}$
	1...0,1	1...0,1	1	0,37...0,99	1	2,70
	1...0,1	1...0,1	1	0,37...0,99	10...1	2,70
	1	1...0,1	1	0,49...0,99	10...1	2,01
	1...0,1	1...0,1	1...0,1	0,49...0,99	10...1	2,01
	1	1...0,1	1...0,1	0,56...0,99	1...10	1,77
	1	1...0,1	1...0,1	0,56...0,99	1...10	1,77
	1...0,1	1	1	0,56...0,99	1...10	1,77
	1	1...0,1	0,1...1	0,79...0,99	0,1...1	1,42
	1...0,1	1	0,1...1	0,73...0,99	1...10	1,35
	1...0,1	1...0,1	1...0,1	0,76...0,99	0,1...1	1,29
	1...0,1	1...0,1	1...0,1	0,76...0,99	0,1...1	1,29
	1...0,1	1...0,1	0,1...1	0,76...0,99	1...0,1	1,29
	1... 10	1...0,1	0,1...1	0,76...0,99	1...10	1,29
	1...0,1	1...0,1	1	0,77...0,99	1...10	1,28
	1...0,1	1...0,1	0,1...1	0,79...0,99	0,1...1	1,26
	1...0,1	1...0,1	1...0,1	0,79...0,99	1...10	1,26
	1...0,1	1...0,1	1...0,1	0,79...0,99	1...10	1,26
	1	1...0,1	1	0,79...0,99	1...10	1,26
	1...0,1	1...0,1	0,1...1	0,79...0,99	10...1	1,26
	1	1...0,1	1	0,81...0,99	1	1,22
	1...0,1	1...0,1	0,1...1	0,86...0,99	1...10	1,16
	1	1...0,1	0,1...1	0,88...0,99	1...10	1,14
	1	1...0,1	0,1...1	0,56...0,99	10...1	0,27

Table 1 was used as a basis for Table 2, in which all the values of the $X4_{\text{suf}} / X4_{\text{sub}}$ ratios are sorted in descending order by groups. This allowed for simplification of selecting a $X4_{\text{su}}$ value depending on a number of variables in review.

**Table 2 – Statistics of variables for X4suf / X4sub
in descending order by groups**

№ п/п	X1 ед.	X2 ед.	X3 ед.	X4 _{su} ед.	S _{su} ² (GDP _{su} ² \$)	X4 _{suf} / X4 _{sub}
1 variable						
	1	1...0,1	1	0,81...0,99	1	1,22
2 variables						
	1...0,1	1...0,1	1	0,37...0,99	1	2,70
	1	1...0,1	1	0,49...0,99	10...1	2,01
	1...0,1	1	1	0,56...0,99	1...10	1,77
	1	1...0,1	1	0,79...0,99	1...10	1,26
3 variables						
	1...0,1	1...0,1	1	0,37...0,99	10...1	2,70
	1	1...0,1	1...0,1	0,56...0,99	1...10	1,77
	1	1...0,1	1...0,1	0,56...0,99	1...10	1,77
	1	1...0,1	0,1...1	0,79...0,99	0,1...1	1,42
	1...0,1	1	0,1...1	0,73...0,99	1...10	1,35
	1...0,1	1...0,1	1	0,77...0,99	1...10	1,28
	1	1...0,1	0,1...1	0,88...0,99	1...10	1,14
	1	1...0,1	0,1...1	0,56...0,99	10...1	0,27
All the variables						
	1...0,1	1...0,1	1...0,1	0,49...0,99	10...1	2,01
	1...0,1	1...0,1	1...0,1	0,76...0,99	0,1...1	1,29
	1...0,1	1...0,1	1...0,1	0,76...0,99	0,1...1	1,29
	1...0,1	1...0,1	0,1...1	0,76...0,99	1...0,1	1,29
	1...10	1...0,1	0,1...1	0,76...0,99	1...10	1,29
	1...0,1	1...0,1	0,1...1	0,79...0,99	0,1...1	1,26
	1...0,1	1...0,1	1...0,1	0,79...0,99	1...10	1,26
	1...0,1	1...0,1	1...0,1	0,79...0,99	1...10	1,26
	1...0,1	1...0,1	0,1...1	0,79...0,99	10...1	1,26
	1...0,1	1...0,1	0,1...1	0,86...0,99	1...10	1,16